

Ex2

A114290041

2026-10-07

```
print(uuid::UUIDgenerate())
```

```
## [1] "163dde87-7377-4979-9db9-150df58e8d0f"
```

1

```
commute <- c(18, 25, 32, 41, 27, 35, 22, 19, 45, 38, 29, 24,  
            31, 36, 40, 26, 33, 28, 21, 44, 30, 37, 23, 39)
```

```
# 1. Mean
```

```
mean(commute)
```

```
## [1] 30.95833
```

```
# 2. Median
```

```
median(commute)
```

```
## [1] 30.5
```

```
# 3. Mode
```

```
table(commute)
```

```
## commute
```

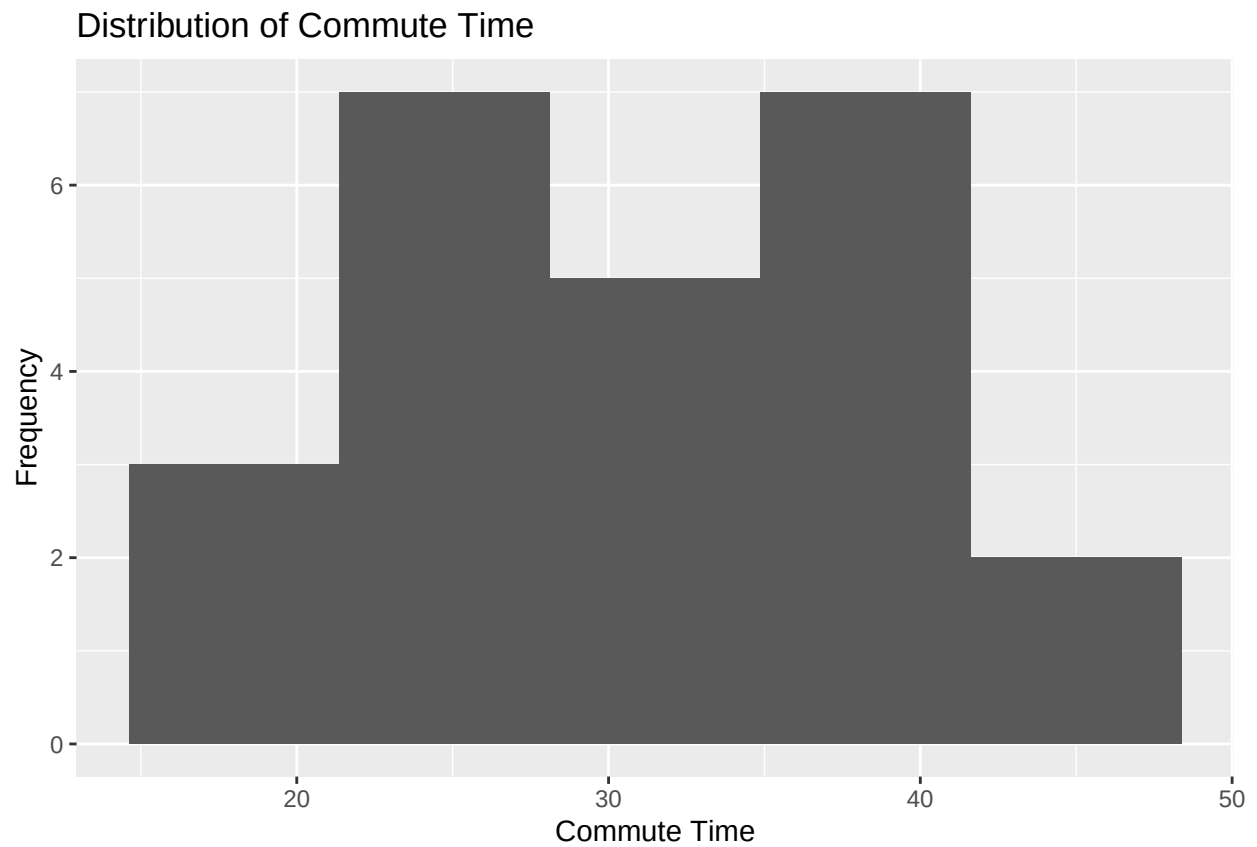
```
## 18 19 21 22 23 24 25 26 27 28 29 30 31 32 33 35 36 37 38 39 40 41 44 45
```

```
##  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
```

2

```
library(ggplot2)
```

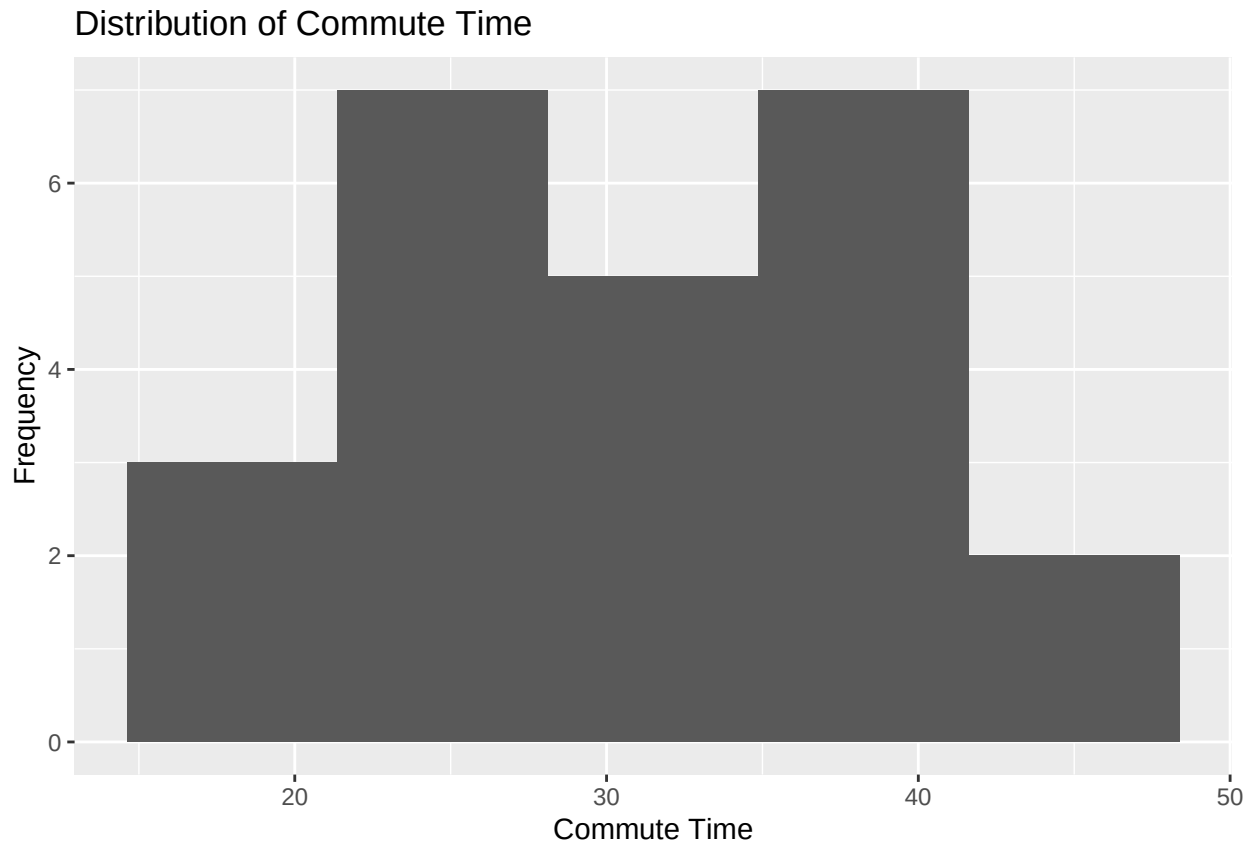
```
ggplot(data.frame(commute), aes(x = commute)) +  
  geom_histogram(bins = 5) +  
  labs(  
    title = "Distribution of Commute Time",  
    x = "Commute Time",  
    y = "Frequency"  
  )
```



```
library(ggplot2)

commute <- c(18, 25, 32, 41, 27, 35, 22, 19, 45, 38, 29, 24,
            31, 36, 40, 26, 33, 28, 21, 44, 30, 37, 23, 39)

ggplot(data.frame(commute), aes(x = commute)) +
  geom_histogram(bins = 5) +
  labs(
    title = "Distribution of Commute Time",
    x = "Commute Time",
    y = "Frequency"
  )
```



```
###3
library(ggplot2)

commute <- c(18, 25, 32, 41, 27, 35, 22, 19, 45, 38, 29, 24,
            31, 36, 40, 26, 33, 28, 21, 44, 30, 37, 23, 39)

# Define 5 intervals
breaks <- seq(15, 45, length.out = 6)

# Calculate cumulative frequency
freq <- hist(commute, breaks = breaks, plot = FALSE)

ogive <- data.frame(
  upper_boundary = freq$breaks[-1],
  cumulative_frequency = cumsum(freq$counts)
)

# Draw the ogive
ggplot(ogive, aes(x = upper_boundary, y = cumulative_frequency)) +
  geom_line() +
  geom_point() +
  labs(
    title = "Ogive of Commute Time",
    x = "Upper Boundary",
    y = "Cumulative Frequency"
  )
```

